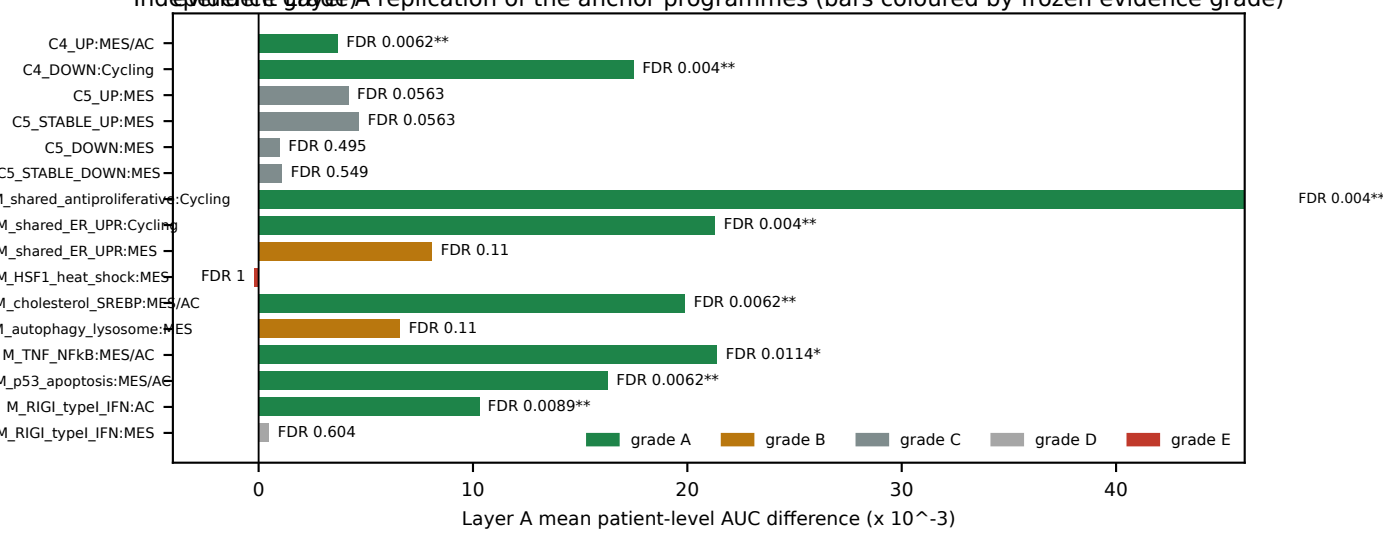
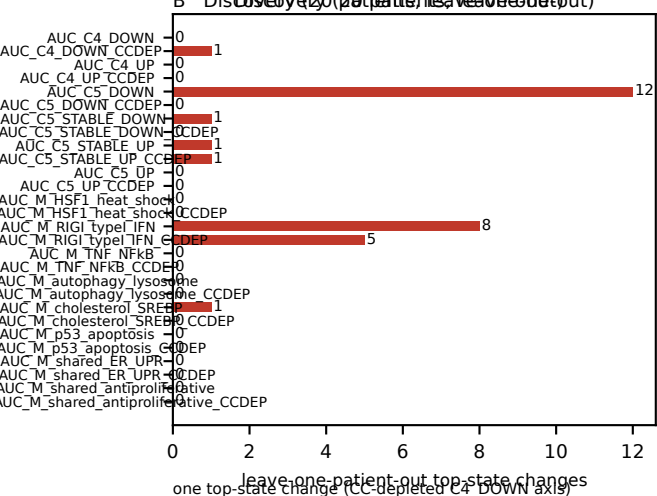


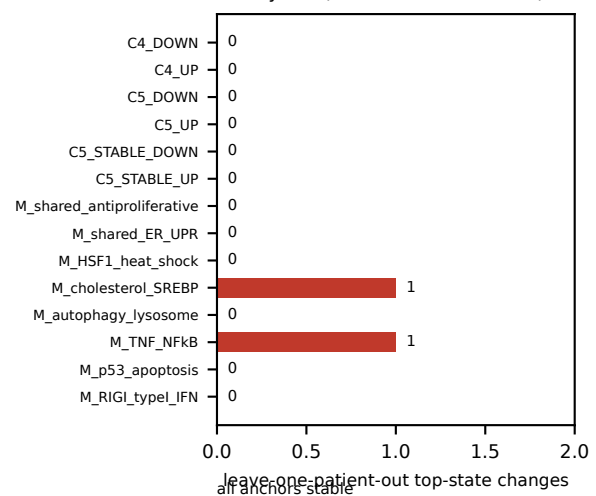
A Independent Layer A replication of the anchor programmes (bars coloured by frozen evidence grade)



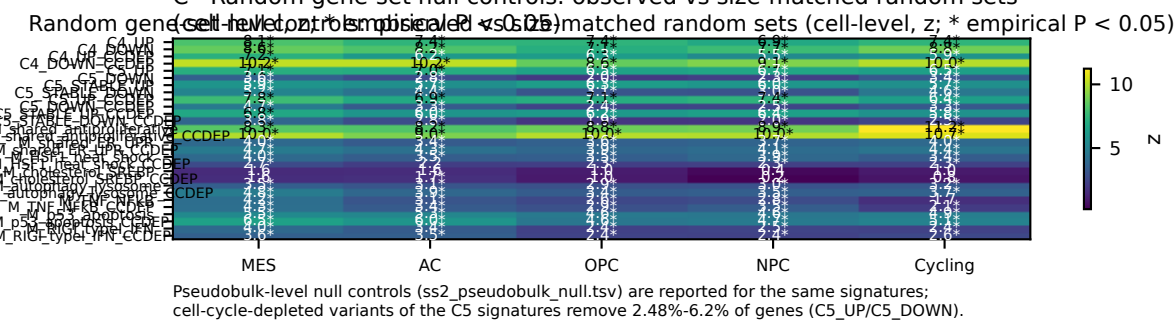
B Discovery (20 patients leave-one-out)



Layer A (12 leave-one-out runs)



C Random gene-set null controls: observed vs size-matched random sets



Pseudobulk-level null controls (ss2_pseudobulk_null.tsv) are reported for the same signatures; cell-cycle-depleted variants of the C5 signatures remove 2.48%-6.2% of genes (C5_UP/C5_DOWN).

D Retained negative findings (findings not removed or removed away)

Retained negative and sensitivity findings carried into the submitted figures:
C4_DOWN and shared anti-proliferative mappings attenuate under expanded-state mapping (FDR 0.594 / 0.249); primary anchors retained with this caveat.
C5 base Layer A MES contrast not significant (FDR 0.0563) -> Grade C, incompletely replicated.
Autophagy-lysosome (MES) not significant in TCGA bulk (n = 44 MES, P = 0.334) -> Grade B sensitivity only.
HSF1 heat-shock reversible between cohorts (Discovery MES FDR 0.002 vs Layer A MES FDR 1.000) -> Grade E, reported as heterogeneity.
CGGA stratification by IDH status and primary/recurrent status does not improve expected-direction recovery (9/21 in every stratum).
Gene-level cross-platform concordance with the comparators is absent (|rho| <= 0.14); shared programmes are a module-level concept only.
C5 does not reproduce on conventional electric-field platforms (gene-level and module-level checks) and remains HITMAN-field-specific.